



AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH

LAB COVER SHEET

Experiment Title:	Study of Different Flip-Flops		
Experiment No:	06	Date of Submission: 9/24/2024	
Course Title:	DIGITAL LOGIC AND CIRCUITS LAB		
Course Code:	COE3103	Section: L	
Semester:	Summer 2023-24	Course Teacher: DR. MD. KABIRUZZAMAN	

Declaration and Statement of Authorship:

1. I/we hold a copy of this Assignment/Case-Study, which can be produced if the original is lost/damaged.
2. This Assignment/Case-Study is my/our original work and no part of it has been copied from any other student's work or from any other source except where due acknowledgement is made.
3. No part of this Assignment/Case-Study has been written for me/us by any other person except where such collaboration has been authorized by the concerned teacher and is clearly acknowledged in the assignment.
4. I/we have not previously submitted or currently submitting this work for any other course/unit.
5. This work may be reproduced, communicated, compared and archived for the purpose of detecting plagiarism.
6. I/we give permission for a copy of my/our marked work to be retained by the faculty for review and comparison, including review by external examiners.
7. I/we understand that Plagiarism is the presentation of the work, idea or creation of another person as though it is your own. It is a form of cheating and is a very serious academic offence that may lead to expulsion from the University. Plagiarized material can be drawn from, and presented in, written, graphic and visual form, including electronic data, and oral presentations. Plagiarism occurs when the origin of the material used is not appropriately cited.
8. I/we also understand that enabling plagiarism is the act of assisting or allowing another person to plagiarize or to copy my/our work.

* Student(s) must complete all details except the faculty use part.

** Please submit all assignments to your course teacher or the office of the concerned teacher.

Group No : 04

No.	Name	ID	Program	Signature
1.	MD.SHAHADAT HOSSAIN	23-50077-1	EEE	Shahadat
2.	TURJOY PAUL	23-50286-1	EEE	Turjoy
3.	TAWKIR ISLAM	22-46249-1	CSE	Tawkir

Faculty use only

FACULTY COMMENTS	Marks Obtained	
	Total Marks	

Title: Study of Different Flip-Flops.

Objective:

To study and understand the design and functionality of various types of flip-flops, including positive edge-triggered D flip-flops (with and without 'Preset' and 'Clear' capabilities), J-K flip-flops, and T flip-flops. Additionally, to design negative edge-triggered flip-flops and a D flip-flop with active high 'Preset' and 'Clear' pins, and to generate corresponding timing diagrams for each flip-flop.

Introduction:

The basic building blocks of combinational logic circuits are gates. In particular, AND, OR, and NOT gates (however, there are also XOR, NAND, NOR, XNOR gates too).

The basic building blocks of sequential logic circuits are flip flops. Flip flops are devices that use a clock. Each flip flop can store one bit. There are different types of flip-flops. D flip-flop, T flip-flop, J-K flip-flop etc.

Theory and Methodology:

Basically, a flip flop has two/three inputs. One input is a control input. For a D flip flop, the control input is labeled D. For a T flip flop, the control input is labelled T. For J-K flipflop the control inputs are J and K. The other input is the clock.

The clock input is usually drawn with a triangular input. These flip-flops are *positive edge-triggered flip flops*. This means that the flip flops can only change output values when the clock is at a positive edge. There are also negative edges triggered flip flops, which change on a negative edge. In this theory section, we consider only positive edge-triggered flip flops.

When the clock is not at a positive edge, then the output value is held. That is, it does not change.

A flip flop also has two outputs, **Q** and **Q'**. The output is really the bit that's stored. Thus, the flip flop is always outputting one bit of information.

But one might wonder "Doesn't it have two bits of information? **Q** and **Q'**?" If we have two bits, we have four possible values. However, **Q'** is the negation of **Q** which means you only have two possible outputs: **Q = 0, Q' = 1** or **Q = 1, Q' = 0**. Since the second output is always negated from the first, you don't get any additional storage. But what is the necessity of the negated output? Actually, the design of flip-flop gives **Q'** basically for free, so that's why flip flops have both the regular output and the negated output.

D Flip-Flop:

In a positive edge triggered D flip-flop, the output **Q** samples the input **D** and becomes **Q = D** only at the positive edge of the clock and it does not change during the whole clock cycle even if the input changes.

Sometimes flip flops often have two additional inputs called *clear* and *preset*. Conventionally they are drawn at the top and bottom of the flip flop respectively. 'Preset' and 'Clear' can be either active low or active high. If the preset is active, the output will be 1 no matter what the conditions of D or clock are. But 'clear' has two types. Asynchronous clear and synchronous clear. If the asynchronous clear is activated, it causes **Q** to be automatically set to 0. It does this, even if the clock has not reached a positive edge. That is, it sets Q to zero as fast as it can. The asynchronous clear is often used to reset flip flops to some initial value. But for synchronous clear, **Q** will be 0 if the clear is active when the clock is in a positive edge.

There are different ways to design a D flip-flop. In this lab sheet, only one way is shown. Students will be familiar about other ways in their theory class.

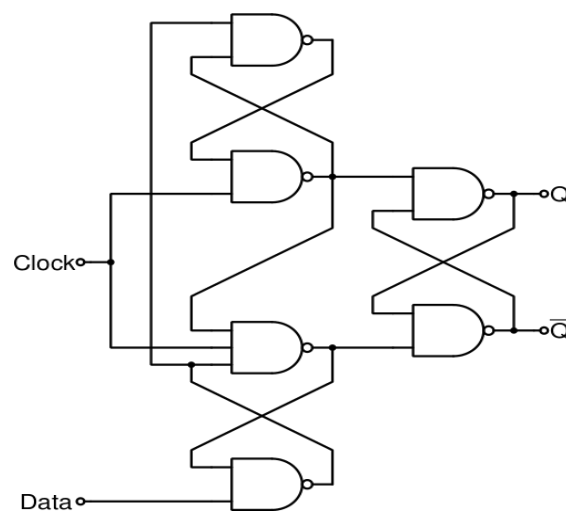


Figure1: Logic circuit a positive edge triggered D flip-flop without preset and clear capability.

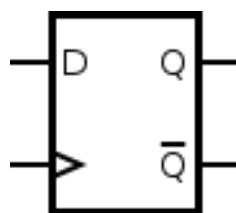


Figure 2: Graphical Symbol of a positive edge triggered D flip-flop without preset and clear capability.

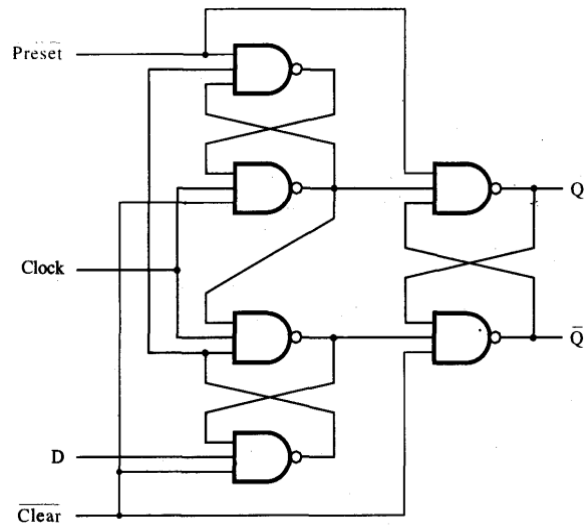
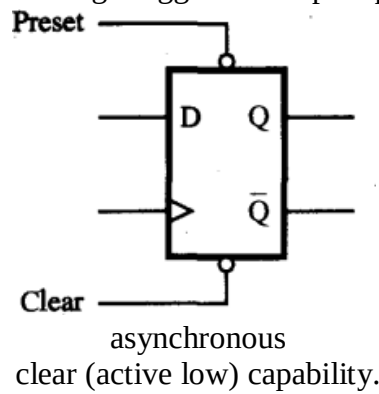


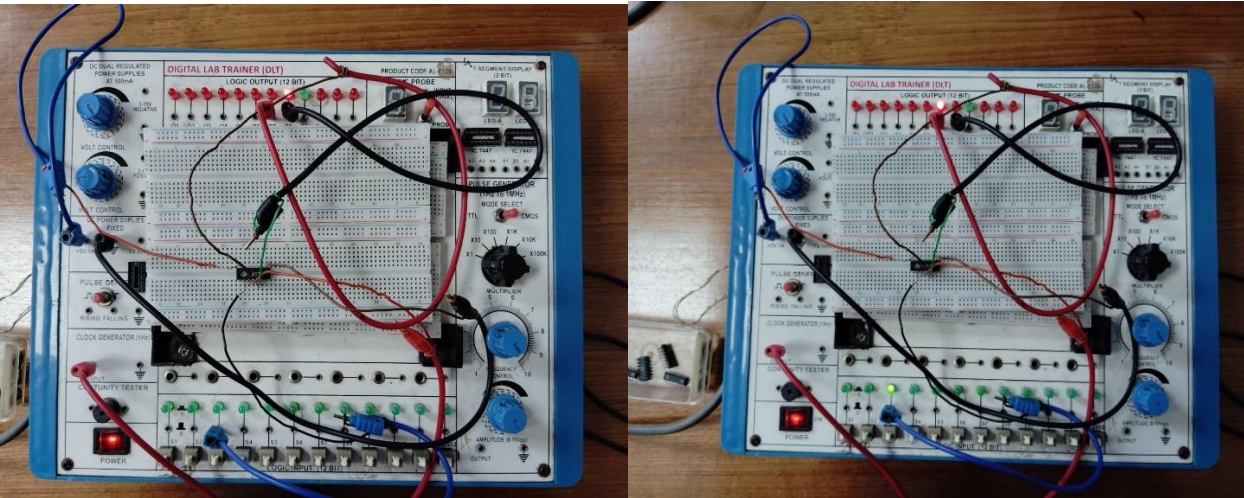
Figure3: Logic circuit a positive edge triggered D flip-flop with preset (active low) and



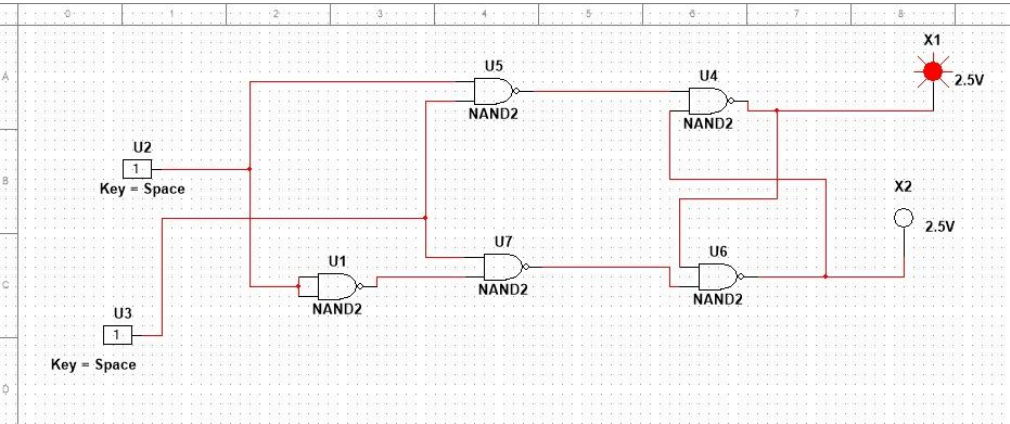
D Flip-flop truth table:

D	Q (Current)	Q(n+1) (Next)
0	0	0
0	1	0
1	0	1
1	1	1

D flip flop:



Simulation:



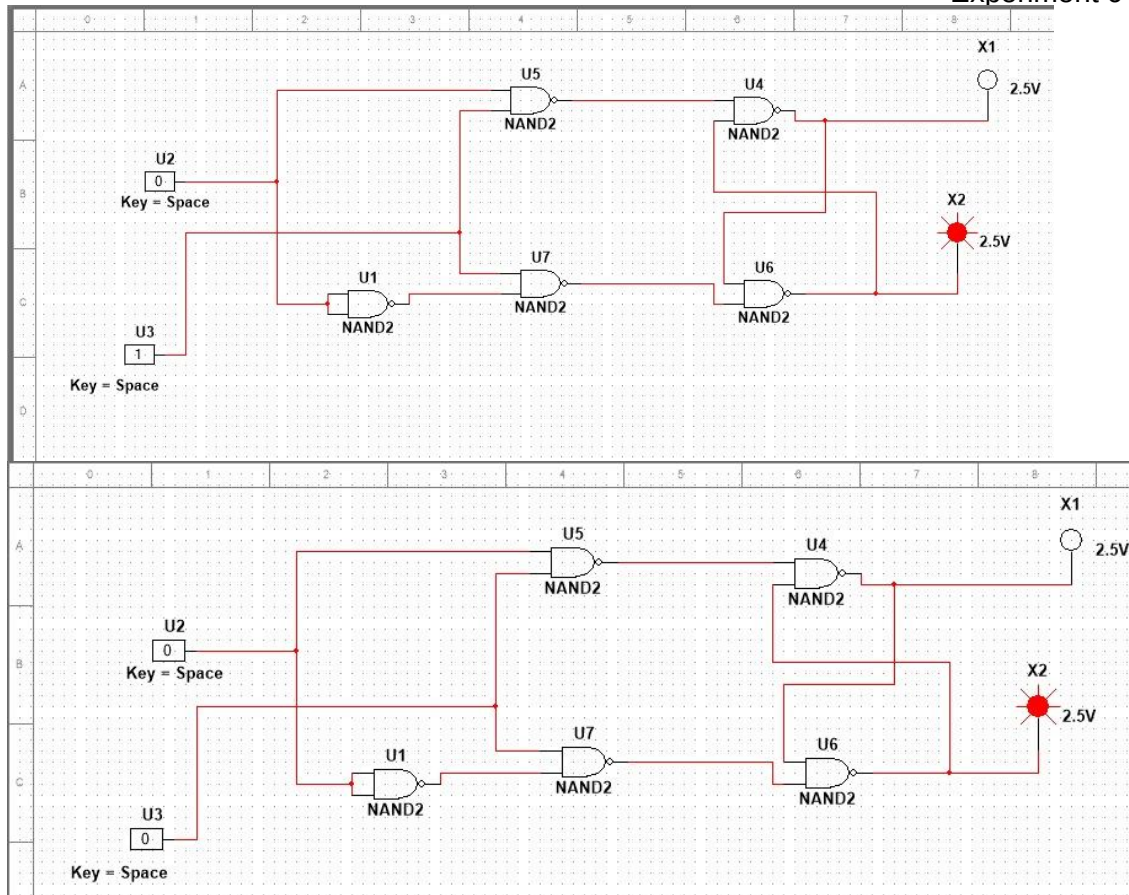
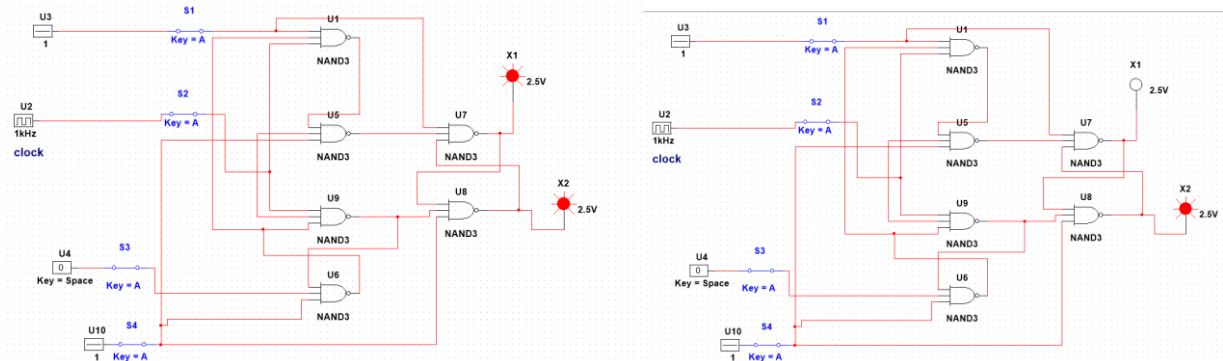


Figure 1: Logic clock circuit a positive edge triggered D flip-flop without preset and clear capability.



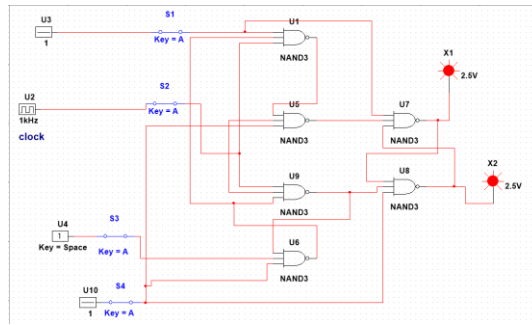


Figure3: Logic circuit a positive edge triggered D flip-flop with preset (active low) and asynchronous clear (active low) capability.

J-K Flip Flop:

In a J-K flip-flop, there are two control input labeled as J and K and a clock input. It has two outputs Q and Q' as usual. In a positive edge triggered J-K flip-flop, output only changes at the positive edge of the clock depending on the values of J and K. If J=1 and K=0, Q is set to 1. If J = 0, K=1 then Q is set to 0. If J = 0 and K = 0 then Q remains unchanged. If J = 1 and K = 1 then Q changes from its former value, which we can say the output toggles.

We can show the characteristics of a J-K in the table given below.

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	$\sim Q(t)$

J-K flip-flops can be designed easily using D flip-flops.

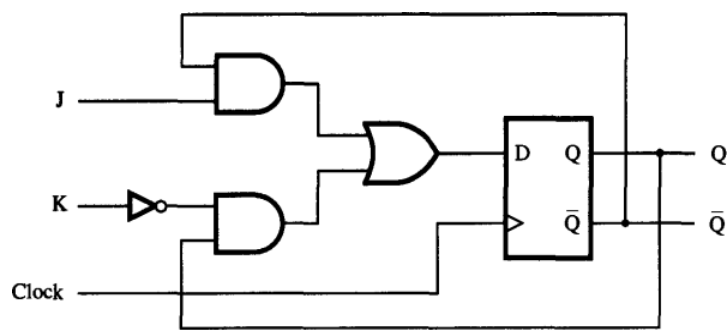


Figure 5: J-K flip-flop using D flip-flop

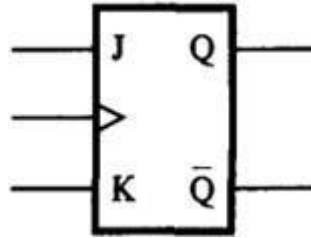
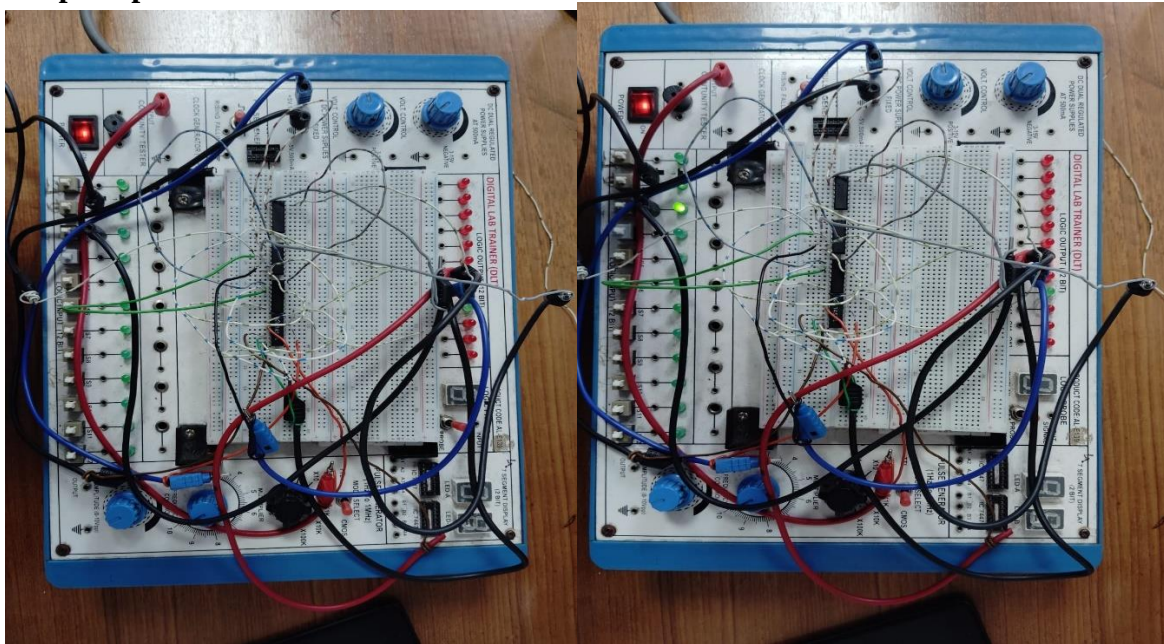


Figure 6: Graphical Symbol of J-K flip-flop.

J-K Flip flop truth table:

J	K	Q_N	Q_{N+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J-K Flip-Flop



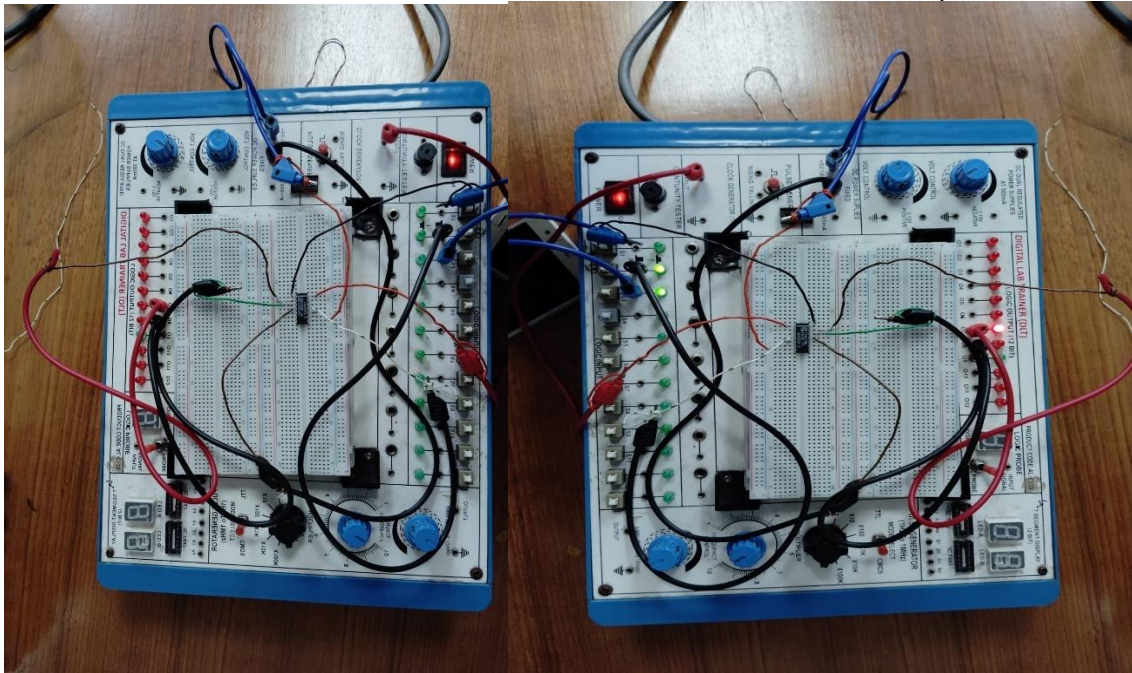
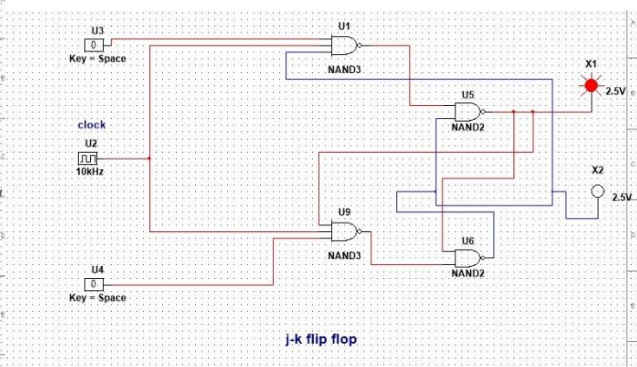
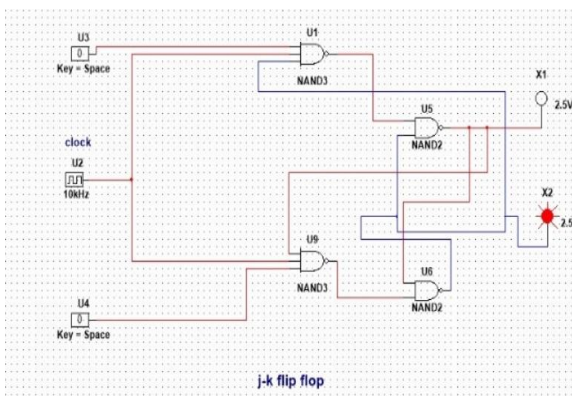
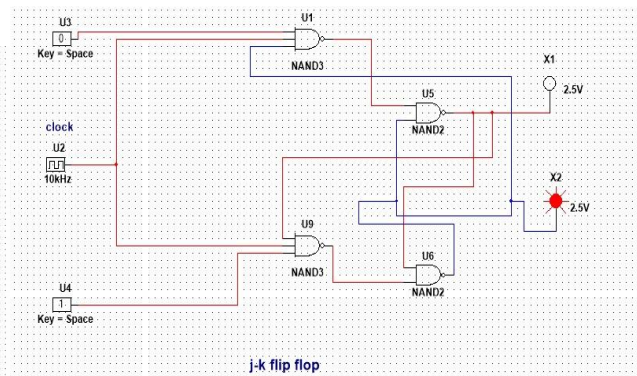
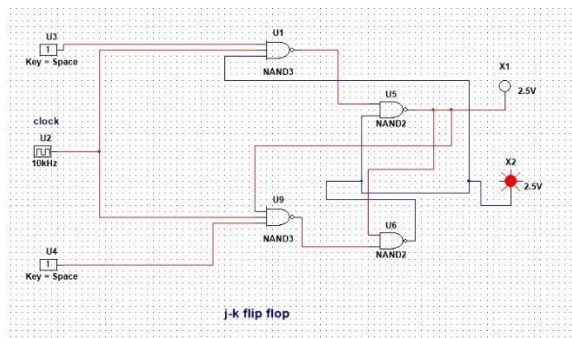


Figure 5: J-K flip-flop using D flip-flop

Simulation:



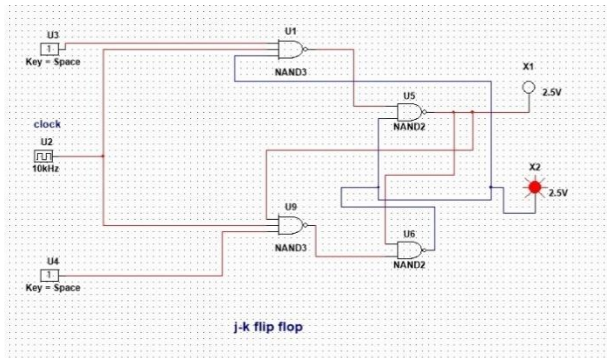


Figure 5: J-K flip-flop using D flip-flop

T Flip-Flop:

A T flip-flop has a control input labeled as T and a clock. The characteristic of a T flip-flop is such that the output toggles at the positive edge of the clock if T is 1. But if T is 0, the output remains unchanged even at the positive edge of the clock.

A T flip-flop can be designed easily by making J and K short of a J-K flip flop.

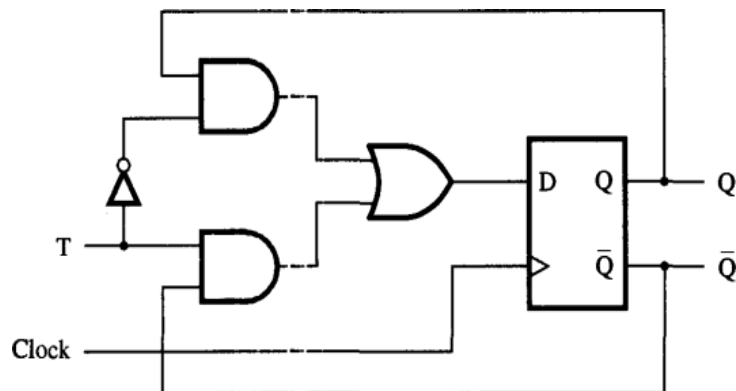


Figure 7: J-K flip-flop using D flip-flop

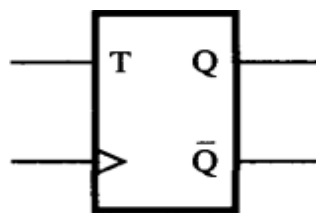


Figure 8: Graphical Symbol of T flip-flop.

There are built-in ICs for D flip-flop and J-K flip-flop. IC-7474 contains 2 D flip-flops and IC-7476 contains 2 J-K flip-flops. The pin configuration of IC-7474 and IC 7476 are given below.

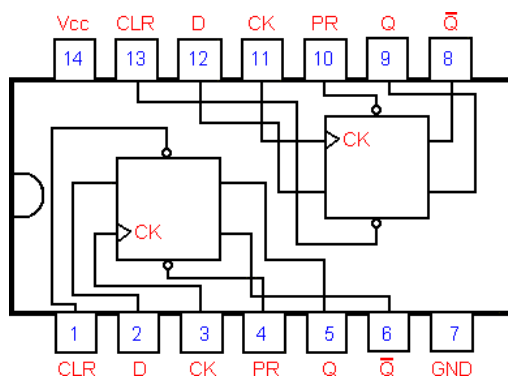


Figure 9: IC-7474

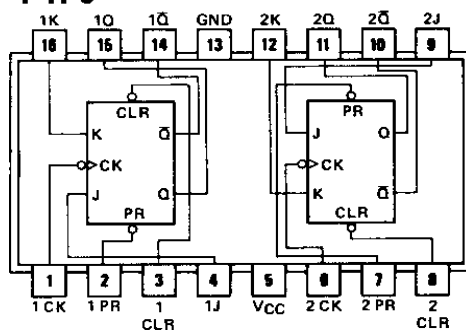
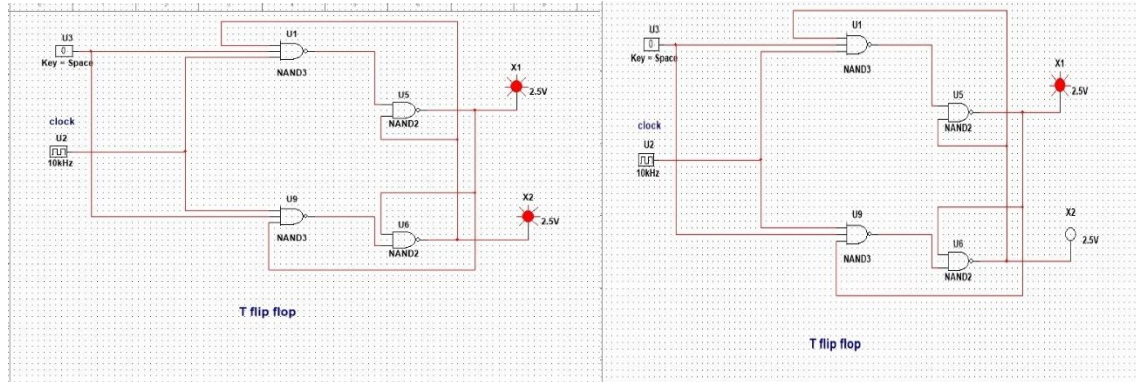
7476

Figure10: IC-7476

T Flip flop truth table:

Q_t	$Q(t+1)$	T
0	0	0
1	0	1
0	1	1
1	1	0

Simulation:**Apparatus:**

- IC: 7404 (NOT Gate) 1[pcs]
 7408 (AND Gate), 2[pcs]
 7432 (OR Gate) 1[pcs]
 7400 (NAND Gate) 6[pcs]
 7474 (D flip-flop) 1[pcs]
 7476 (J-K flip-flop) 1[pcs]

Experimental Procedure:

The circuits shown in Figure 1, Figure 3, Figure 5, and Figure 7 were implemented on the trainer board, and the input-output characteristics were observed. A pulse switch was used as the clock. Timing diagrams were generated from the implemented circuits.

Simulation and Measurement:

Timing Diagram of a positive edge triggered D flip-flop without 'Preset' and 'Clear' capability:

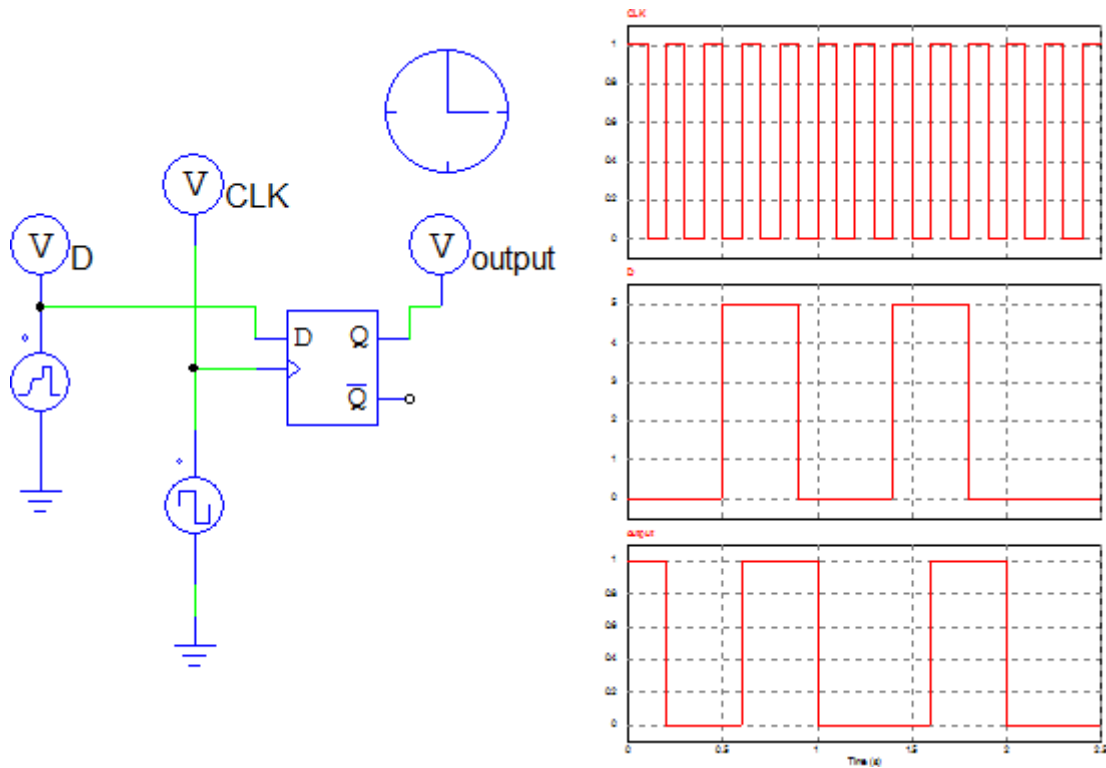


Figure11: Timing Diagram of a positive edge triggered D Flip-Flop without preset and clear capability. Output only samples the input when the clock has a positive edge and remains unchanged for the next Timing Diagram of a positive edge triggered J-K flip-flop:

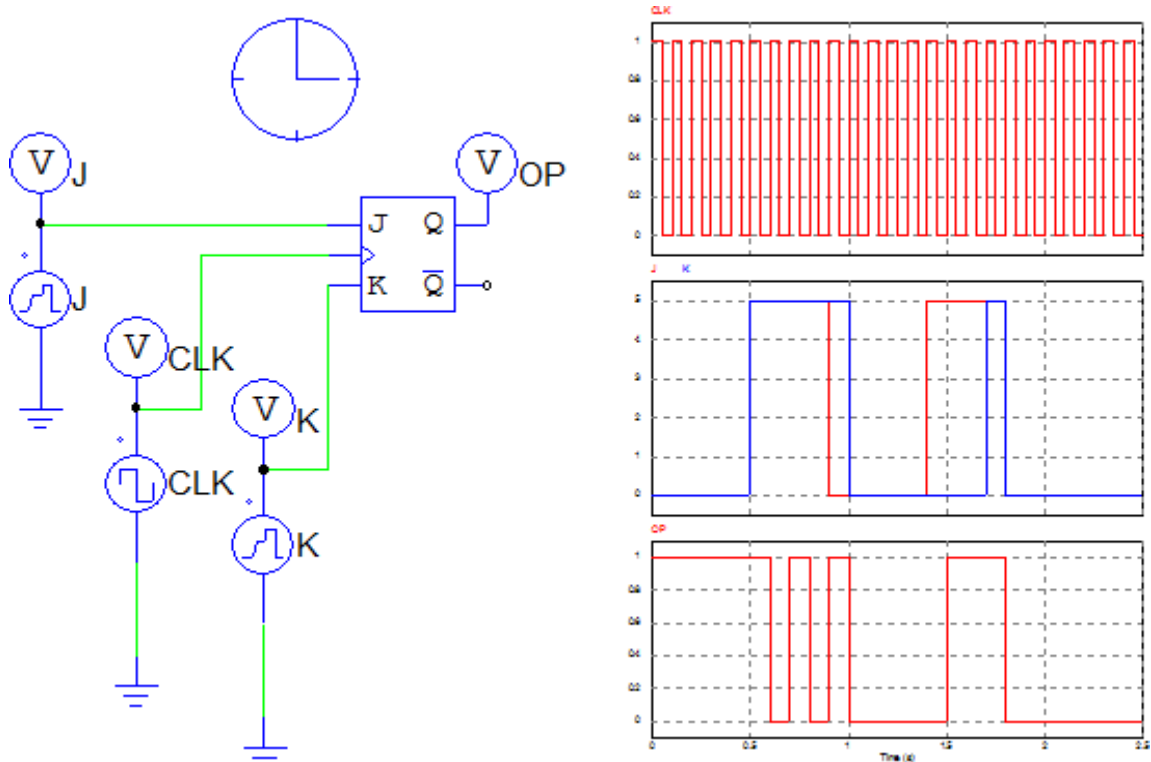


Figure12: Timing Diagram of J-K Flip-Flop Timing Diagram of a positive edge triggered T flip-flop:

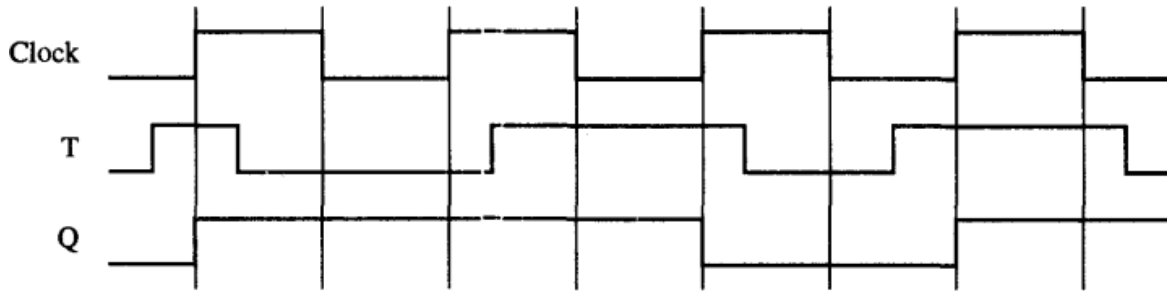


Figure13: Timing Diagram of T-flip-flop

Discussion and Conclusion:

After this experiment was performed, students became familiar with the design of positive edge-triggered D flip-flops with and without 'Preset' and 'Clear' capabilities, J-K flip-flops, and T flip-flops. Following the experiment, their task was to design negative edge-triggered flip-flops and to draw the timing diagrams for them. They were also required to design and draw the timing diagram for a D flip-flop with an active high 'Preset' and 'Clear' pin. Positive edge.

Reference:

1. "Fundamentals of Digital Logic with Verilog design" by – Brown & Vranesic
2. www.wikipedia.org
3. <http://www.cs.umd.edu/class/spring2003/cmsc311/Notes/Seq/flip.html>